

Apply filters to SQL queries

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Project description

As a security professional, my role involves investigating suspicious activity within the organization's systems to maintain a secure environment. This project focuses on examining potential security issues by analyzing data from the `employees` and `log_in_attempts` tables. The goal is to use SQL queries with appropriate filters to retrieve and review records related to failed login attempts and access patterns. By identifying failed logins, especially those occurring after business hours or from unexpected employee accounts, I can help uncover signs of unauthorized access or compromised credentials and take necessary steps to mitigate risks.

Retrieve after hours failed login attempts

This SQL query is used to investigate a potential security incident by identifying all **failed login attempts** that occurred **after business hours** (after 18:00). This helps in monitoring suspicious activity that may have taken place outside normal working times.

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [organization]> clear
MariaDB [organization]> clear
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_time > '18:00' AND success = 0;

+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0 |
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 | 0 |
| 20 | tahah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 | 0 |
| 28 | aestrada | 2022-05-09 | 19:28:12 | MEXICO | 192.168.27.57 | 0 |
| 34 | drocas | 2022-05-11 | 21:02:04 | US | 192.168.45.93 | 0 |
| 42 | cgriffin | 2022-05-09 | 23:04:05 | US | 192.168.4.157 | 0 |
| 52 | cjaackson | 2022-05-10 | 22:07:07 | CAN | 192.168.58.57 | 0 |
| 69 | wjaaffrey | 2022-05-11 | 19:55:15 | USA | 192.168.100.17 | 0 |
| 82 | abernard | 2022-05-12 | 23:38:46 | MEX | 192.168.234.49 | 0 |
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 | 0 |
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 | 0 |
| 104 | asundara | 2022-05-11 | 18:38:07 | US | 192.168.96.200 | 0 |
| 107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0 |
| 111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0 |
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0 |
| 131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0 |
| 155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0 |
| 160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0 |
| 199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0 |
+-----+-----+-----+-----+-----+-----+-----+

19 rows in set (0.131 sec)

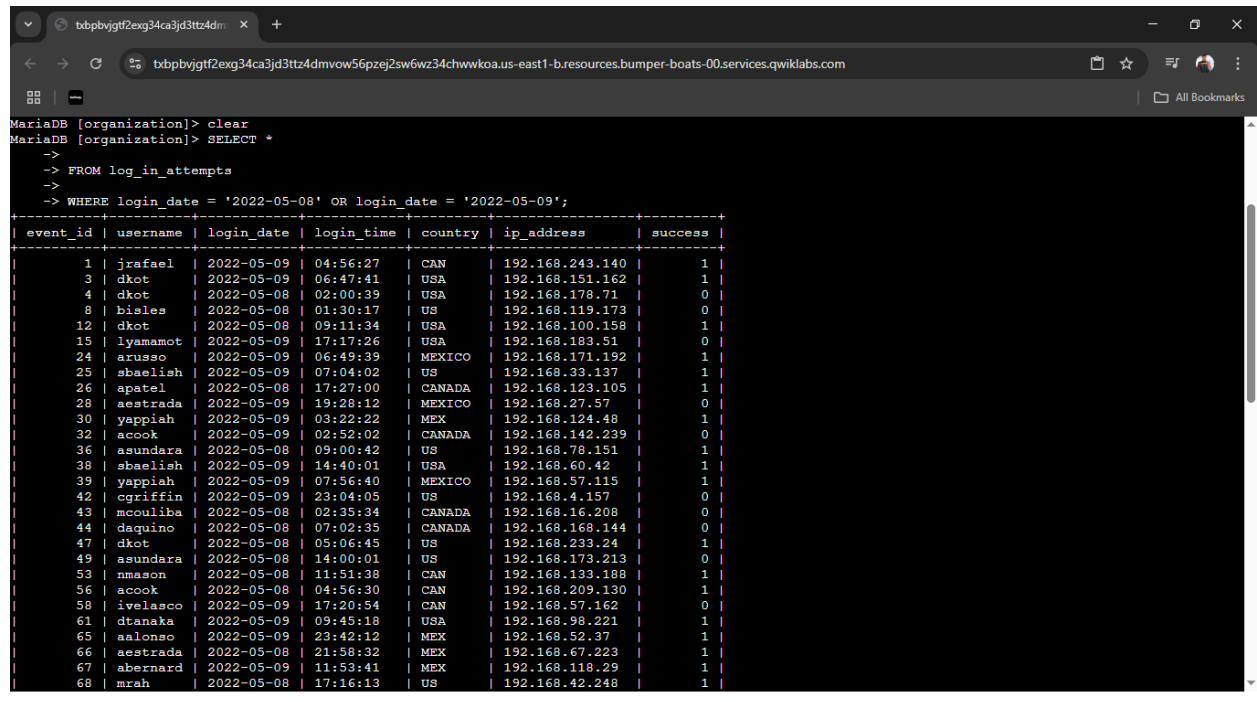
MariaDB [organization]> 
```

This query retrieves all records from the `log_in_attempts` table where login attempts failed (`success = 0`) and occurred after 6:00 PM (`TIME(login_time) > '18:00:00'`).

- `TIME(login_time)` extracts only the time portion of the `login_time` column (assuming it's stored as a `DATETIME` or `TIMESTAMP`).
- `'18:00:00'` represents 6:00 PM in 24-hour format.
- `success = 0` filters for login attempts that were not successful.

Retrieve login attempts on specific dates

A suspicious event happened on 2022-05-09, so I wrote this query to review all login attempts from that day and the day before. The query selects all records from the `log_in_attempts` table where the `login_date` is either `'2022-05-08'` or `'2022-05-09'`. This helps me see any unusual login activity leading up to the incident.



```

MariaDB [organization]> clear
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date = '2022-05-08' OR login_date = '2022-05-09';

```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.131.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	abaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	abaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrah	2022-05-08	17:16:13	US	192.168.42.248	1

Retrieve login attempts outside of Mexico

I wrote this query to pull every login attempt that didn't come from Mexico. By using **WHERE NOT country LIKE 'MEX%'**, it excludes any record where the **country** value starts with "MEX" (covering both "MEX" and "MEXICO"). This way, I see only those attempts originating outside of Mexico for my investigation.

txbpbvjgtf2exg34ca3jd3ttz4dm

txbpbvjgtf2exg34ca3jd3ttz4dmvw56pje2sw6wz34chwwkoa.us-east1-b:resources.bumper-boats-00.services.qwiklabs.com

MariaDB [organization]> SELECT *

->

-> FROM log_in_attempts

->

-> WHERE NOT country LIKE 'MEX';

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.131.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgillmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	arrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	abaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyanamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusao	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	abaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1

Retrieve employees in Marketing

I wrote this query to fetch every employee who works in the Marketing department and is located in the East building. The condition `department = 'Marketing'` filters for Marketing staff, and `office LIKE 'East%'` matches any office code that starts with “East” (such as East-170 or East-320). This lets me target exactly those machines in the East building’s Marketing team for security updates.

```
txbpbvjgtf2exg34ca3jd3ttz4dm: x +
txbpbvjgtf2exg34ca3jd3ttz4dmvow56pzej2sw6wz34chwwkoa.us-east1-b:resources.bumper-boats-00.services.qwiklabs.com
All Bookmarks

1183 | b566c710d544 | lquraish | Human Resources | East-400 |
1184 | c986d200e170 | ptsosie | Human Resources | Central-247 |
1185 | d790e839f461 | revens | Sales | North-330 |
1186 | e281f433g404 | sacosta | Sales | North-460 |
1187 | f963g637h851 | bbode | Finance | East-351 |
1188 | g164h566i795 | noshiro | Finance | West-252 |
1189 | h784i120j837 | slefkowi | Human Resources | West-342 |
1190 | NULL | kcarter | Marketing | Central-270 |
1191 | NULL | shakimi | Marketing | Central-366 |
1192 | k570l183m949 | rlaghari | Information Technology | East-138 |
1193 | l186m618n319 | esantiag | Information Technology | Central-300 |
1194 | m340n287o441 | zwarren | Human Resources | West-212 |
1195 | n516o853p957 | orainier | Finance | East-346 |
1196 | o22ep357q829 | sshah2 | Information Technology | South-385 |
1197 | p791q114r509 | aabara | Information Technology | North-159 |
1198 | q308r573s459 | jmartine | Marketing | South-117 |
1199 | r520s571t459 | areyes | Human Resources | East-100 |
-----
200 rows in set (0.001 sec)

MariaDB [organization]> SELECT * FROM emoloyees WHERE department = 'Marketing' AND office LIKE 'East%';
ERROR 1146 (42S02): Table 'organization.emoloyees' doesn't exist
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
-----
| employee_id | device_id | username | department | office |
-----
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautiat | Marketing | East-267 |
| 1088 | k865l965m233 | rgoash | Marketing | East-157 |
| 1103 | NULL | randerss | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
-----
7 rows in set (0.001 sec)

MariaDB [organization]>
```

Retrieve employees in Finance or Sales

I wrote this query to pull every employee who works in either the Finance or Sales department. By using `WHERE department IN ('Finance', 'Sales')`, it filters the `employees` table to return only records where the `department` value matches “Finance” or “Sales.” This ensures I get the complete list of staff in those two departments for the security update.

txbpbvjgtf2exg34ca3jd3ttz4dm: X +

txbpbvjgtf2exg34ca3jd3ttz4dmvw56pzej2sw6wz34chwwkoa.us-east1-b:resources.bumper-boats-00.services.qwiklabs.com

All Bookmarks

```

MariaDB [organization]> SELECT * FROM employees WHERE department = 'Finance' OR 'Sales';

```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1010	k242l212m542	jlansky	Finance	South-109
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1029	d336e475f676	ivelasco	Finance	East-156
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1062	k367l639m697	redwards	Finance	North-180
1069	NULL	jpark	Finance	East-110
1076	y347z204a710	fgarcia	Finance	Central-270
1081	d647e310f618	qcorbit	Finance	South-290
1083	f840g812h544	gkoshi	Finance	West-165
1099	v283w690x104	anaser	Finance	West-357
1105	b551c837d758	kmei	Finance	Central-232
1122	s103t952u851	btorres	Finance	West-319
1136	g299h520i457	jhawes	Finance	West-416
1138	i671j355k725	xromero	Finance	South-329
1142	m674n127o823	lsilva	Finance	East-440
1144	NULL	erobinso	Finance	Central-266
1147	r454s225t299	tvega	Finance	West-177
1148	s328t505u907	dharvey	Finance	South-181
1159	d881e710f732	jshen	Finance	East-193
1164	i682j513k442	fsmeltz	Finance	North-163

Retrieve all employees not in IT

This query fetches every employee record except those in the Information Technology department. The `WHERE NOT department = 'Information Technology'` clause filters out IT staff, returning all other departments so the update can be applied only to non-IT machines.

txbpbvjgtf2xg34ca3jd3ttz4dmvw56pzej2sw6wz34chwwkoa.us-east1-b:resources.bumper-boats-00.services.qwiklabs.com

MariaDB [organization]> `SELECT * FROM employees WHERE NOT department = 'Information Technology';`

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrdrriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jaoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550o824t230	jclark	Finance	North-188
1018	s310u840w653	abellmas	Finance	North-403
1020	u899w381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	biales	Sales	South-171
1036	k550l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1039	n253o917p623	cjackson	Sales	East-378
1040	o783p832q294	dtarly	Human Resources	East-237
1041	p929q222r778	cgriffin	Sales	North-208

Summary

I wrote queries to retrieve failed login attempts after 18:00, on May 8 and 9, and originating outside of Mexico, helping to trace unauthorized access without sifting through all records. I also filtered the `employees` table to isolate staff in Marketing (East building), in Finance or Sales, and everyone except IT, ensuring updates are applied only where needed. Together, these SQL filters support precise incident response and efficient rollout of security updates.